

Emmanuel Asiimwe

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Profile

Computational Engineering student at Aalto University with production backend experience: co-founded Cadera Solutions and led the backend of its school information system. Most interested in verification: knowing that a system does what it claims.

Experience

Co-founder & Backend Lead, Cadera Solutions Ltd (formerly Videra Digital) Jun 2025 – Jun 2026

Cadera, a school information system covering students, classes, grading and reporting; in production at one school since December 2025.

- Designed the PostgreSQL data model and NestJS service layer, including the business logic for grading, records and reporting
- Implemented authentication and role-based access control
- Built the report-generation pipeline and record archiving
- Owned deployment end to end

Lead Web Developer, Business Initiatives International Limited

Mar 2024 – Jul 2024

- Built the company's first website in React

Projects

PolyTrend

Jun 2023 – May 2024

Polynomial regression and model-selection tool (Python, NumPy/SciPy, Qt GUI; GPL-3.0)

- Fits polynomial models by least squares and selects the degree by Bayesian Information Criterion, reporting R^2 , AIC/BIC and residual diagnostics
- Tested with pytest using deterministic, version-controlled fixtures; packaged with uv and PyInstaller
- Built for and distributed to a 50-student IB science cohort

VoteAble · Able Corp (student startup)

Oct 2023 – Jan 2024

Real-time election results web app (React, Bootstrap, Bun)

- Used for the school's student elections; time from opening of voting to published results fell from 2+ days to 8 hours

Education

Aalto University, BSc in Computational Engineering – Espoo, Finland

2026 – 2029 (expected)

Aga Khan High School, IB Diploma in Mathematics AA, Physics, Digital Society (HL) –
Kampala, Uganda

Sep 2023 – Jul 2025

- Score: 39/45

Skills

Languages: TypeScript, Python, JavaScript, C++

Backend: NestJS, Node.js, PostgreSQL, Prisma, REST API design

Frontend: React, Qt (PyQt)

Scientific computing: NumPy, SciPy, Matplotlib, pytest, PyTorch

Tools: Linux, Git, Docker, uv